

Chapter 6.9

Hyperbolic Functions and Hanging Cables

Introduction

We can write

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} = C + S$$

where C is an even function (**hyperbolic cosine of x**) and S is an odd function (**hyperbolic sine of x**).

Euler's Formula: $e^{ix} = \cos x + i \sin x$ and $e^{-ix} = \cos x - i \sin x$

So that $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ and $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

Now we can write

$$(i) \quad \cos(ix) = \frac{e^{i(ix)} + e^{i(-ix)}}{2} = \frac{e^{-x} + e^x}{2} = \cosh x$$

$$(ii) \quad \sin(ix) = \frac{e^{i(ix)} - e^{i(-ix)}}{2i} = \frac{e^{-x} - e^x}{2i} = -i \frac{e^x - e^{-x}}{2} = -i \sinh x$$

De Moivre's Formula: $(e^{ix})^n = (\cos x + i \sin x)^n = \cos(nx) + i \sin(nx)$

6.9.1 Definition

Hyperbolic sine: $\sinh x = \frac{e^x - e^{-x}}{2}$

Hyperbolic cosine: $\cosh x = \frac{e^x + e^{-x}}{2}$

Hyperbolic tangent: $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

Hyperbolic cotangent: $\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Hyperbolic secant: $\operatorname{sech} x = \frac{2}{e^x + e^{-x}}$

Hyperbolic cosecant: $\operatorname{csch} x = \frac{2}{e^x - e^{-x}}$

Example 1 We have $\sinh x = \frac{e^x - e^{-x}}{2}$

So $\sinh 0 = \frac{e^0 - e^{-0}}{2} = \frac{1-1}{2} = 0$, $\cosh 0 = \frac{e^0 + e^{-0}}{2} = \frac{1+1}{2} = 1$, $\sinh 2 = \frac{e^2 - e^{-2}}{2} \approx 3.6269$

Graphs of hyperbolic functions

Hanging cables and applications

1. Vibrating motions inside elastic solids.
2. Flexible cable suspended between two points, a telephone line hanging between two poles.
3. Catenary: The cable has an equation $y = a \cosh\left(\frac{x}{a}\right) + c$ where a is the distance between the poles.

Hyperbolic identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csc} h^2 x$$

6.9.2 Theorem

Basic Formulas

$\cosh x + \sinh x = e^x$	$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$
$\cosh x - \sinh x = e^{-x}$	$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$
$\cosh^2 x - \sinh^2 x = 1$	$\sinh(x - y) = \sinh x \cosh y - \cosh x \sinh y$
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$\cosh(x - y) = \cosh x \cosh y - \sinh x \sinh y$
$\coth^2 x - 1 = \operatorname{csc} h^2 x$	$\sinh 2x = 2 \sinh x \cosh x$
$\cosh(-x) = \cosh x$	$\cosh 2x = \cosh^2 x + \sinh^2 x$
$\sinh(-x) = -\sinh x$	$\cosh 2x = 2 \sinh^2 x + 1 = 2 \cosh^2 x - 1$

Derivative and Integral Formulas

$\frac{d}{dx}(\sinh x) = \cosh x$	$\int \sinh u \, du = \cosh u + c$
$\frac{d}{dx}(\cosh x) = \sinh x$	$\int \cosh u \, du = \sinh u + c$
$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$	$\int \operatorname{sech}^2 u \, du = \tanh u + c$
$\frac{d}{dx}(\coth x) = -\operatorname{csc} h^2 x$	$\int \operatorname{csc} h^2 u \, du = -\coth u + c$
$\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$	$\int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u + c$
$\frac{d}{dx}(\operatorname{csc} h x) = -\operatorname{csc} h x \coth x$	$\int \operatorname{csc} h u \coth u \, du = -\operatorname{csc} h u + c$

6.9.3 Theorem

Generalized Derivative Formulas

$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$	$\frac{d}{dx}(\coth u) = -\operatorname{csc} h^2 u \frac{du}{dx}$
$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$	$\frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$
$\frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{du}{dx}$	$\frac{d}{dx}(\operatorname{csc} h u) = -\operatorname{csc} h u \coth u \frac{du}{dx}$

Example 2

$$\frac{d}{dx}[\cosh(x^3)] = \sinh(x^3) \frac{d}{dx}(x^3) = 3x^2 \sinh(x^3)$$

Formula: $\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$ where $u = x^3$

$$\frac{d}{dx}[\ln(\tanh x)] = \frac{1}{\tanh x} \frac{d}{dx}(\tanh x) = \frac{\operatorname{sech}^2 x}{\tanh x}$$

Formula: $\frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{du}{dx}$ where $u = x$

Example 3

$$\int \sinh^5 x \cosh x \, dx = \frac{1}{6} \sinh^6 x + c \quad \text{Here } u = \sinh x \Rightarrow du = \cosh x \, dx$$

$$\int \tanh x \, dx = \int \frac{\sinh x}{\cosh x} \, dx = \ln|\cosh x| + c = \ln(\cosh x) + c \quad \text{because } \cosh x > 0$$

Here $u = \cosh x \Rightarrow du = \sinh x \, dx$

Example 4

A 100 ft wire is attached at its ends to the tops of two 50 ft poles that are positioned 90 ft apart. How high above the ground is the middle of the wire?

Solution: Please see [Section 6.4](#).

6.9.4 Theorem

Relation between inverse hyperbolic and logarithmic functions

(Logarithmic Form of Inverse Hyperbolic Functions)

$$\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$$

$$\cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$$

$$\tanh^{-1}(x) = \frac{1}{2} \ln \frac{1+x}{1-x}$$

$$\coth^{-1}(x) = \frac{1}{2} \ln \frac{x+1}{x-1}$$

$$\operatorname{sech}^{-1}(x) = \ln \frac{1 + \sqrt{1 - x^2}}{x}$$

$$\operatorname{csch}^{-1}(x) = \ln \left(\frac{1}{x} + \frac{\sqrt{1 + x^2}}{|x|} \right)$$

Example 5

$$\sinh^{-1} 1 = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2}) \approx 0,8814$$

$$\tanh^{-1}(1/2) = \frac{1}{2} \ln \left(\frac{1 + (1/2)}{1 - (1/2)} \right) = \frac{1}{2} \ln 3 \approx 0,5493$$

Derivatives Involving Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2 + 1}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}} \text{ for } x > 1$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1 - x^2} \text{ for } |x| < 1$$

$$\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1 - x^2} \text{ for } |x| > 1$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1 - x^2}} \text{ for } 0 < x < 1$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1 + x^2}} \text{ for } x \neq 0$$

6.9.5 Theorem

Generalized Derivative Formulas

$$\frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{u^2 + 1}} \frac{du}{dx}$$

$$\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx} \text{ for } u > 1$$

$$\frac{d}{dx}(\tanh^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx} \text{ for } |u| < 1$$

$$\frac{d}{dx}(\coth^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx} \text{ for } |u| > 1$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} u) = -\frac{1}{u\sqrt{1 - u^2}} \text{ for } 0 < u < 1$$

$$\frac{d}{dx}(\operatorname{csch}^{-1} u) = -\frac{1}{|u|\sqrt{1 + u^2}} \text{ for } u \neq 0$$

6.9.6 Theorem

Integrals Involving Inverse Hyperbolic Functions

$$\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1} \left(\frac{u}{a} \right) + c = \ln(u + \sqrt{u^2 + a^2}) + c$$

In particular, $\int \frac{du}{\sqrt{x^2 + 1}} = \sinh^{-1} x + c = \ln(x + \sqrt{x^2 + 1}) + c$

$$\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1} \left(\frac{u}{a} \right) + c = \ln(u + \sqrt{u^2 - a^2}) + c \text{ for } u > a$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \begin{cases} \frac{1}{a} \tanh^{-1} \left(\frac{u}{a} \right) + c, & |u| < a \\ \frac{1}{a} \coth^{-1} \left(\frac{u}{a} \right) + c, & |u| > a \end{cases}$$
$$= \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right| + c, \quad |u| \neq a$$

$$\int \frac{du}{u\sqrt{a^2 - u^2}} = -\frac{1}{a} \operatorname{sech}^{-1} \left| \frac{u}{a} \right| + c = -\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - u^2}}{|u|} \right) + c, \quad 0 < |u| < a$$

$$\int \frac{du}{u\sqrt{a^2 + u^2}} = -\frac{1}{a} \operatorname{csch}^{-1} \left| \frac{u}{a} \right| + c = -\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 + u^2}}{|u|} \right) + c, \quad u \neq 0$$

Example 6

Evaluate $\int \frac{dx}{\sqrt{4x^2 - 9}}, x > \frac{3}{2}$.

Solution: Let $u = 2x$. Then $du = 2 dx$. Therefore, we get

$$\begin{aligned}\int \frac{dx}{\sqrt{4x^2 - 9}} &= \frac{1}{2} \int \frac{2dx}{\sqrt{4x^2 - 9}} = \frac{1}{2} \int \frac{du}{\sqrt{u^2 - 9}} \\ &= \frac{1}{2} \cosh^{-1}\left(\frac{u}{3}\right) + c = \frac{1}{2} \cosh^{-1}\left(\frac{2x}{3}\right) + c \\ &= \frac{1}{2} \ln(2x + \sqrt{4x^2 - 9}) + c\end{aligned}$$

EXERCISE SET 6.9

9. $y = \sinh(4x - 8)$

$$\frac{dy}{dx} = 4 \cosh(4x - 8)$$

Formula: $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$ where $u = 4x - 8$

10. $y = \cosh(x^4)$

$$\frac{dy}{dx} = 4x^3 \sinh(x^4)$$

Formula: $\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$ where $u = x^4$

11. $y = \coth(\ln x)$

$$\frac{dy}{dx} = -\frac{1}{x} \operatorname{csc} h^2(\ln x)$$

Formula: $\frac{d}{dx}(\coth u) = -\operatorname{csc} h^2 u \frac{du}{dx}$ where $u = \ln x$

12. $y = \ln(\tanh 2x)$

$$\frac{dy}{dx} = \frac{1}{\tanh 2x} (\sec h^2 2x)(2) = \frac{2 \sec h^2 2x}{\tanh 2x}$$

Formula: $\frac{d}{dx}(\tanh u) = \sec h^2 u \frac{du}{dx}$ where $u = 2x$

13. $y = \operatorname{csc} h(1/x)$

$$\frac{dy}{dx} = -\operatorname{csc} h(1/x) \coth((1/x) \left(-\frac{1}{x^2}\right) = \frac{1}{x^2} \operatorname{csc} h(1/x) \coth((1/x)$$

Formula: $\frac{d}{dx}(\operatorname{csc} h u) = -\operatorname{csc} h u \coth u \frac{du}{dx}$ where $u = \frac{1}{x}$

14. $y = \operatorname{sech}(e^{2x})$

$$\frac{dy}{dx} = -\operatorname{sech}(e^{2x}) \tanh(e^{2x}) \times e^{2x} \times 2 = -2e^{2x} \operatorname{sech}(e^{2x}) \tanh(e^{2x})$$

Formula: $\frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$ where $u = e^{2x}$

15. $y = \sqrt{4x + \cosh^2(5x)}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{4x + \cosh^2(5x)}} [4 + 2 \cosh(5x) \sinh(5x) \times 5] = \frac{4 + 5 \sinh(10x)}{2\sqrt{4x + \cosh^2(5x)}}$$

16. $y = \sinh^3(2x)$

$$\begin{aligned} \frac{dy}{dx} &= 3\sinh^2(2x) \cosh(2x) \times 2 \\ &= 6\sinh^2(2x) \cosh(2x) \end{aligned}$$

17. $y = x^3 \tanh^2(\sqrt{x})$

$$\begin{aligned} \frac{dy}{dx} &= x^3 \left[2 \tanh(\sqrt{x}) \operatorname{sech}^2(\sqrt{x}) \left(\frac{1}{2\sqrt{x}} \right) \right] + 3x^2 \tanh^2(\sqrt{x}) \\ &= x^{5/2} \tanh(\sqrt{x}) \operatorname{sech}^2(\sqrt{x}) + 3x^2 \tanh^2(\sqrt{x}) \end{aligned}$$

18. $y = \sinh(\cos 3x)$

$$\begin{aligned} \frac{dy}{dx} &= \cosh(\cos 3x) (-3 \sin 3x) \\ &= -3 \cosh(\cos 3x) \sin 3x \end{aligned}$$

20. $y = \sinh^{-1}(1/x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\sqrt{1 + \left(\frac{1}{x}\right)^2}} \left(-\frac{1}{x^2} \right) \\ &= -\frac{1}{x\sqrt{x^2 + 1}} \end{aligned}$$

Formula: $\frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{1+u^2}} \frac{du}{dx}$ where $u = \frac{1}{x}$

22. $y = \cosh^{-1}(\sinh^{-1} x)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{(\sinh^{-1} x)^2 - 1}} \times \frac{1}{\sqrt{1+x^2}}$$

Formula: $\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}$ where $u = \sinh^{-1} x$

24. $y = (\coth^{-1} x)^2$

$$\frac{dy}{dx} = 2 \coth^{-1} x \times \frac{1}{1-x^2}$$

Formula: $\frac{d}{dx}(\coth^{-1} u) = \frac{1}{1-u^2} \frac{du}{dx}$

25. $y = \cosh^{-1}(\cosh x)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 x - 1}} \sinh x = \frac{\sinh x}{\sqrt{\sinh^2 x}} = \frac{\sinh x}{|\sinh x|}$$

Formula: $\frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}$ where $u = \cosh x$

26. $y = \sinh^{-1}(\tanh x)$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 + \tanh^2 x}} \sec h^2 x$$

Formula: $\frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{1+u^2}} \frac{du}{dx}$ where $u = \tanh x$

27. $y = e^x \sec h^{-1} \sqrt{x}$

$$\begin{aligned} \frac{dy}{dx} &= e^x \frac{-1}{\sqrt{x} \sqrt{1 - (\sqrt{x})^2}} \times \frac{1}{2\sqrt{x}} + e^x \sec h^{-1} \sqrt{x} \\ &= \frac{-e^x}{2x\sqrt{1-x}} + e^x \sec h^{-1} \sqrt{x} \end{aligned}$$

Formula: $\frac{d}{dx}(\sec h^{-1} u) = -\frac{1}{u\sqrt{1-u^2}} \frac{du}{dx}$ where $u = \sqrt{x}$

28. $y = (1 + x \csc h^{-1} x)^{10}$

$$\frac{dy}{dx} = 10(1 + x \csc h^{-1} x)^9 \left[x \times \frac{-1}{|x|\sqrt{1-x^2}} + \csc h^{-1} x \right]$$

Formula: $\frac{d}{dx}(\csc h^{-1} u) = -\frac{1}{|u|\sqrt{1+u^2}} \frac{du}{dx}$

Evaluate the following integrals:

29. $\int \sinh^6 x \cosh x \, dx$

Here $u = \sinh x \Rightarrow du = \cosh x \, dx$

$$\int \sinh^6 x \cosh x \, dx = \int u^6 \, du$$

$$= \frac{u^7}{7} + c = \frac{1}{7} \sinh^7 x + c$$

Formula: $\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$

34. $\int \coth^2 x \csc h^2 x dx$ Here $u = \coth x \Rightarrow du = -\csc h^2 x dx$

$$\int \coth^2 x \csc h^2 x dx = -\int u^2 du$$

$$= -\frac{u^3}{3} + c = -\frac{1}{3} \coth^3 x + c$$

Formula: $\frac{d}{dx}(\coth u) = -\csc h^2 u \frac{du}{dx}$

35. $\int_{\ln 2}^{\ln 3} \tanh x \sec h^3 x dx$ Here $u = \sec h x \Rightarrow du = -\sec h x \tanh x dx$

$$\int_{\ln 2}^{\ln 3} \tanh x \sec h^3 x dx = -\int_{\ln 2}^{\ln 3} \sec h^2 x (-\sec h x \tanh x) dx$$

$$= -\frac{1}{3} [\sec h^3 x]_{\ln 2}^{\ln 3} = -\frac{1}{3} [\sec h^3 (\ln 3) - \sec h^3 (\ln 2)]$$

Formula: $\frac{d}{dx}(\sec h u) = -\sec h u \tanh u \frac{du}{dx}$

36. $\int_0^{\ln 3} \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$ Here $u = e^x + e^{-x} \Rightarrow du = (e^x - e^{-x}) dx$

$$\int_0^{\ln 3} \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \int_0^{\ln 3} \frac{du}{u} = [\ln(u)]_0^{\ln 3}$$

$$= \left[\ln\left(3 + \frac{1}{3}\right) - \ln(1+1) \right] = \ln \frac{10}{3} - \ln 2 = \ln \frac{5}{3}$$

38. $\int \frac{dx}{\sqrt{x^2 - 2}}, x > \sqrt{2}$ Here ~~$u = e^x + e^{-x} \Rightarrow du = (e^x - e^{-x}) dx$~~

$$\int \frac{dx}{\sqrt{x^2 - 2}} = \int \frac{dx}{\sqrt{x^2 - (\sqrt{2})^2}}$$

$$= \cosh^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$

$$= \ln(x + \sqrt{x^2 - 2}) + c$$

Formula: $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1}\left(\frac{u}{a}\right) + c = \ln(u + \sqrt{u^2 - a^2}) + c$ for $u > a$, where $a = \sqrt{2}$

40. $\int \frac{\sin \theta}{\sqrt{1 + \cos^2 \theta}} d\theta$ Here $u = \cos \theta \Rightarrow du = -\sin \theta d\theta$

$$\int \frac{\sin \theta}{\sqrt{1 + \cos^2 \theta}} d\theta = -\int \frac{du}{\sqrt{1 + u^2}} = -\sinh^{-1} u + c$$

$$= -\sinh^{-1}(\cos \theta) + c$$

$$= \ln(\cos \theta + \sqrt{1 + \cos^2 \theta}) + c$$

Formula: $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1}\left(\frac{u}{a}\right) + c = \ln(u + \sqrt{u^2 + a^2}) + c$ where $u = \cos \theta$

42. $\int \frac{dx}{\sqrt{9x^2 - 25}}, x > \frac{5}{3}$

$$\begin{aligned} \int \frac{dx}{\sqrt{9x^2 - 25}} &= \int \frac{dx}{\sqrt{9\left(x^2 - \frac{25}{9}\right)}} = \frac{1}{3} \int \frac{dx}{\sqrt{x^2 - (5/3)^2}} \\ &= \frac{1}{3} \cosh^{-1}\left(\frac{x}{5/3}\right) + c \\ &= \frac{1}{3} \ln\left(x + \sqrt{x^2 - (5/3)^2}\right) + c \\ &= \left\{ \begin{aligned} &\frac{1}{3} \cosh^{-1}\left(\frac{3x}{5}\right) + c \\ &\ln\left(x + \frac{1}{3}\sqrt{9x^2 - 25}\right) + c \end{aligned} \right. \end{aligned}$$

Formula: $\int \frac{du}{\sqrt{u^2 - a^2}} = \cosh^{-1}\left(\frac{u}{a}\right) + c = \ln(u + \sqrt{u^2 - a^2}) + c$ for $u > a$, where $a = \frac{5}{3}$

44. $\int_0^{\sqrt{3}} \frac{dt}{\sqrt{t^2 + 1}}$

$$\begin{aligned} \int_0^{\sqrt{3}} \frac{dt}{\sqrt{t^2 + 1}} &= \left[\sinh^{-1} t \right]_0^{\sqrt{3}} \\ &= \left[\ln(t + \sqrt{t^2 + 1}) \right]_0^{\sqrt{3}} \\ &= \ln(\sqrt{3} + 2) \end{aligned}$$

Formula: $\int \frac{du}{\sqrt{a^2 + u^2}} = \sinh^{-1}\left(\frac{u}{a}\right) + c = \ln(u + \sqrt{u^2 + a^2}) + c$ where $u = t$ and $a = 1$